INTERNSHIP PROJECT REPORT

Project Title: Smart Farmer Crop Advisor Using Python Full Stack with Django Framework
Student: Dadimi Yasashwini
College: Annamacharya Institute of Technology & Sciences, Rajampet
Department: Computer Science and Engineering
Internship Company: DevSprint Services Pvt. Ltd.
Duration: 22 May 2026 – 10 July 2026

1.Acknowledgement

I would like to express my sincere gratitude to **DevSprint Services Pvt. Ltd.** for providing me with an opportunity to undertake this internship project. The internship enabled me to gain practical knowledge in Python Full Stack Development using the Django Framework and understand how modern web applications are designed and implemented.

I express my heartfelt thanks to the management and faculty members of the Department of Computer Science and Engineering, Annamacharya Institute of Technology and Sciences, Rajampet, for their continuous encouragement and valuable guidance throughout the internship.

I also thank my friends and family members for their constant support and motivation during the development of this project.

Finally, I thank everyone who directly or indirectly contributed to the successful completion of this internship project.

2.Abstract

Agriculture is one of the most important sectors contributing to the economic development of India. Farmers often face challenges in selecting suitable crops, estimating water requirements, maintaining cultivation records, and calculating expected profits. Traditional methods involve manual calculations and paper-based record keeping, which are time-consuming and prone to errors.

The **Smart Farmer Crop & Irrigation Advisor System** is a web-based application developed using the **Python Django Framework** to provide a simple and effective digital solution for these challenges. The application allows users to register securely, log in, and access various agricultural utilities through a user-friendly dashboard.

The project consists of modules such as Crop Recommendation, Irrigation Calculator, Farmer Records Management, and Profit Estimator. These modules assist users in selecting crops based on soil type and season, estimating irrigation requirements, maintaining farmer records digitally, and calculating expected profits based on land area and crop type.

The application is developed using **Python**, **Django**, **HTML**, **CSS**, **JavaScript**, and **SQLite**. Django's Model-View-Template (MVT) architecture ensures organized code structure, secure authentication, and efficient database management.

The proposed system demonstrates how modern web technologies can simplify agricultural management by providing quick calculations, secure data storage, and an easy-to-use interface. It also serves as an educational project to understand Django Full Stack Development and can be enhanced in the future with Artificial Intelligence, weather forecasting, and mobile application support.

3.INTRODUCTION

Agriculture is considered the backbone of India's economy as it provides employment and food security to a large population. Farmers are responsible for producing crops that meet the country's growing demand. However, many agricultural activities are still performed manually, making farm management difficult and time-consuming.

With the advancement of Information Technology, web applications have become powerful tools for solving real-world problems. A web-based agricultural management system helps farmers organize their farming activities, maintain digital records, estimate irrigation requirements, and calculate expected profits with greater accuracy.

The **Smart Farmer Crop & Irrigation Advisor System** has been developed using the Django Framework to provide an integrated platform for farmers. The application includes user authentication, crop recommendation, irrigation calculation, farmer record management, and profit estimation.

The system provides a clean and responsive interface that allows users to access different services from a centralized dashboard. All farmer records are stored in an SQLite database, ensuring efficient data management and easy retrieval.

The project demonstrates the practical implementation of Python Full Stack Development using Django and highlights how software technology can contribute to modern agriculture.

4.Problem Statement

Agriculture is one of the most important occupations in India, but many farmers still depend on traditional methods for crop selection, irrigation planning, and maintaining cultivation records. Most of these activities are carried out manually, which consumes time and increases the possibility of errors. Farmers often find it difficult to estimate the amount of water required for irrigation and the expected profit from cultivation. In addition, maintaining records on paper makes it difficult to retrieve information when needed.

To overcome these challenges, there is a need for a simple and user-friendly digital system that allows farmers to manage their farming activities efficiently. The proposed **Smart Farmer Crop & Irrigation Advisor System** provides a centralized platform where users can securely log in, obtain crop recommendations, estimate irrigation requirements, maintain farmer records, and calculate expected profits.

5.Objectives

The main objectives of the project are:

- To develop a secure web application using the Django framework.
- To provide user registration and authentication.
- To recommend suitable crops based on soil type and season.
- To calculate irrigation water requirements based on crop type and land area.
- To maintain farmer information digitally using a database.
- To estimate the expected profit from cultivation.
- To provide a simple, attractive, and user-friendly interface.
- To reduce paperwork and improve data management.

6.Existing System

In the existing system, most farming activities are managed manually. Farmers maintain records using notebooks or paper documents, making it difficult to store, retrieve, and update information. Crop selection is generally based on traditional knowledge rather than structured recommendations, and irrigation planning often relies on experience rather than calculated estimates. Manual profit estimation can also lead to inaccurate planning.

Disadvantages of Existing System

- Manual record keeping.
- Time-consuming calculations.
- Higher chances of errors.
- No centralized database.
- Difficult to maintain historical records.
- Limited accessibility of information.

7.Proposed System

The proposed **Smart Farmer Crop & Irrigation Advisor System** is developed using Python and the Django Framework. The application provides a secure login system, allowing registered users to access agricultural services through a single dashboard. The system stores information in an SQLite database, making data retrieval fast and reliable.

The application includes modules for crop recommendation, irrigation calculation, farmer record management, and profit estimation. These features simplify farming-related activities and reduce manual effort.

Advantages of the Proposed System

- Secure user authentication.
- Digital record management.
- Quick crop recommendation.
- Accurate irrigation estimation.
- Easy profit calculation.
- User-friendly interface.
- Efficient database management.

8. Software Requirements

Software	Description
Operating System	Windows 10 / Windows 11
Programming Language	Python 3.x
Framework	Django
Frontend	HTML5, CSS3, JavaScript
Database	SQLite
IDE	Visual Studio Code

Software	Description
Browser	Google Chrome / Microsoft Edge

9. Hardware Requirements

Hardware	Specification
Processor	Intel Core i3 or above
RAM	Minimum 4 GB
Storage	500 MB Free Space
Internet	Required for development and updates
Display	1366 × 768 Resolution or higher

10.Technologies Used

The **Smart Farmer Crop & Irrigation Advisor System** was developed using modern web technologies that provide a secure, scalable, and user-friendly platform. The technologies selected ensure efficient development, easy maintenance, and reliable performance.

Python

Python is the primary programming language used for developing the application. It is widely used because of its simple syntax, readability, and extensive libraries. Python provides powerful backend support and integrates seamlessly with the Django framework. It allows developers to write clean and maintainable code while reducing development time.

Django Framework

Django is a high-level Python web framework that follows the **Model-View-Template (MVT)** architecture. It provides built-in features such as user authentication, URL routing, database management, and security against common web attacks. Django simplifies web application development by reducing repetitive coding and improving project organization.

HTML

HTML (HyperText Markup Language) is used to create the structure of web pages. It defines the layout of forms, buttons, tables, and navigation menus used throughout the project.

CSS

CSS (Cascading Style Sheets) is used to improve the appearance of the application. It provides colors, fonts, spacing, animations, responsive layouts, and modern UI components, making the application visually attractive and user-friendly.

JavaScript

JavaScript is used to provide client-side validation and interactive features. It improves user experience by validating form inputs before submission and adding dynamic behaviour to web pages.

SQLite

SQLite is the database used in this project. It is lightweight, serverless, and easy to integrate with Django. The database stores user accounts, farmer profiles, and farmer records securely.

Visual Studio Code

Visual Studio Code (VS Code) is the Integrated Development Environment (IDE) used to develop the application. It provides intelligent code completion, debugging tools, terminal support, and extensions for Python and Django development.

11. System Architecture

The Smart Farmer Crop & Irrigation Advisor System follows the **Django Model-View-Template (MVT)** architecture.

The user accesses the application through a web browser. User requests are handled by Django URLs, which forward them to the appropriate views. The views process the business logic and interact with database models whenever necessary. Finally, the processed information is displayed to the user using HTML templates.

Architecture Flow

```
User
  │
  ▼
Web Browser
  │
  ▼
Django URLs
  │
  ▼
Views (Business Logic)
  │
  ▼
Models
  │
  ▼
SQLite Database
  │
  ▼
Templates (HTML/CSS/JS)
  │
  ▼
User Interface.
```

12. Module Description

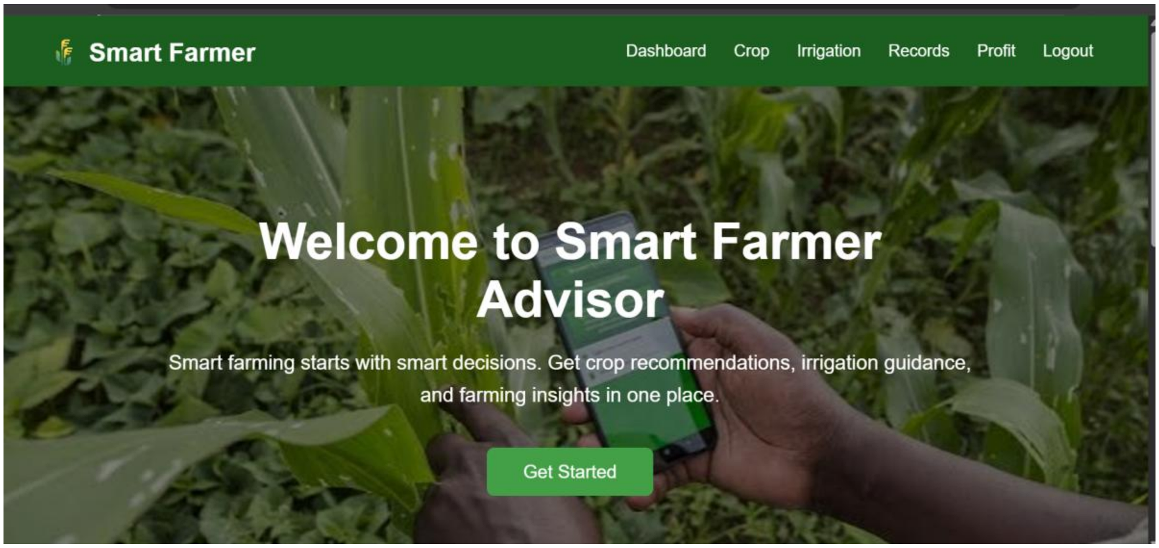
This section explains each module of the application in detail.

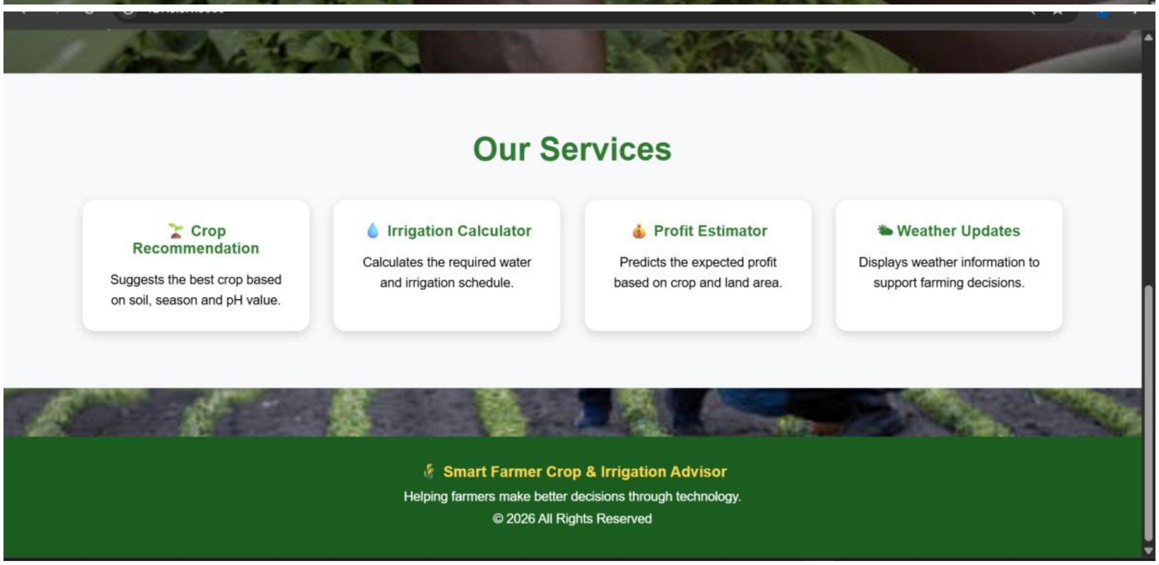
12.1 Home Module

The Home Module is the main entry point of the application. It provides an introduction to the Smart Farmer Crop & Irrigation Advisor System and allows users to navigate to the Login and Registration pages. The page is designed with an agriculture-themed interface to provide a professional appearance and improve user experience.

Features

- Attractive homepage
- Navigation bar
- Background image
- Responsive design
- Footer section



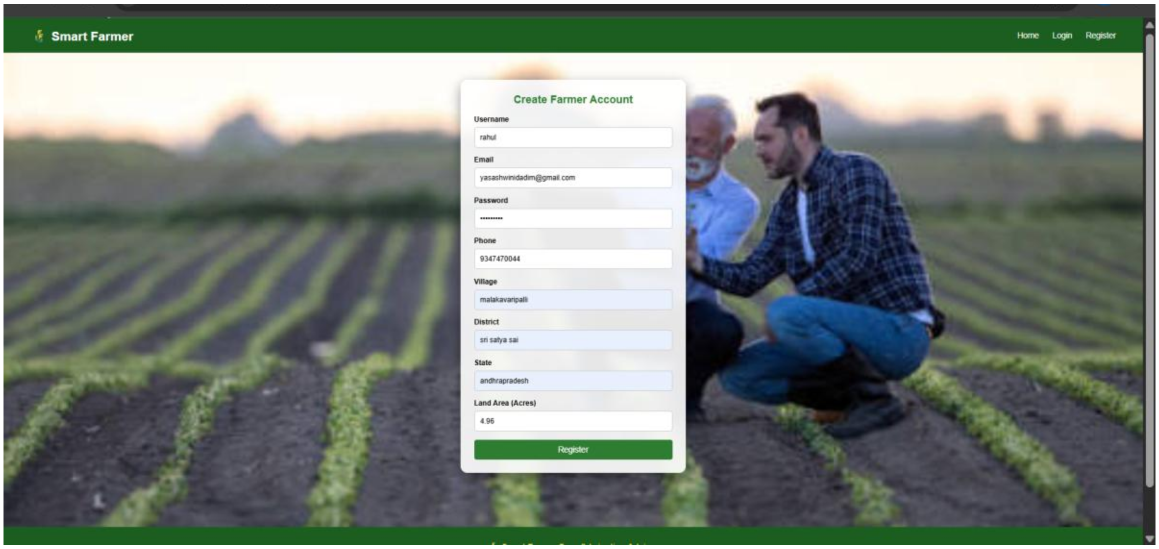


12.2 Registration Module

The Registration Module allows new users to create an account by entering their personal and farmer-related information. The system validates the entered data before storing it in the SQLite database. Duplicate usernames are not allowed, ensuring unique user accounts.

Features

- User registration
- Form validation
- Secure password storage

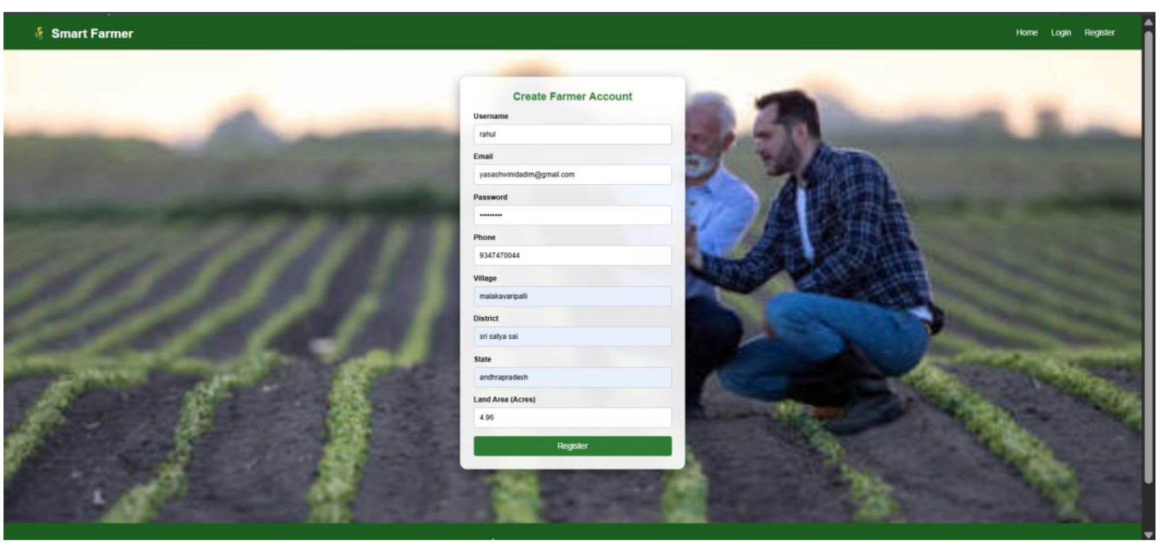


12.3 Login Module

The Login Module authenticates registered users using their username and password. Django's authentication system verifies the credentials and redirects authenticated users to the dashboard. Invalid login attempts display appropriate error messages.

Features

- Secure authentication
- Error handling
- Session management
- Dashboard redirection



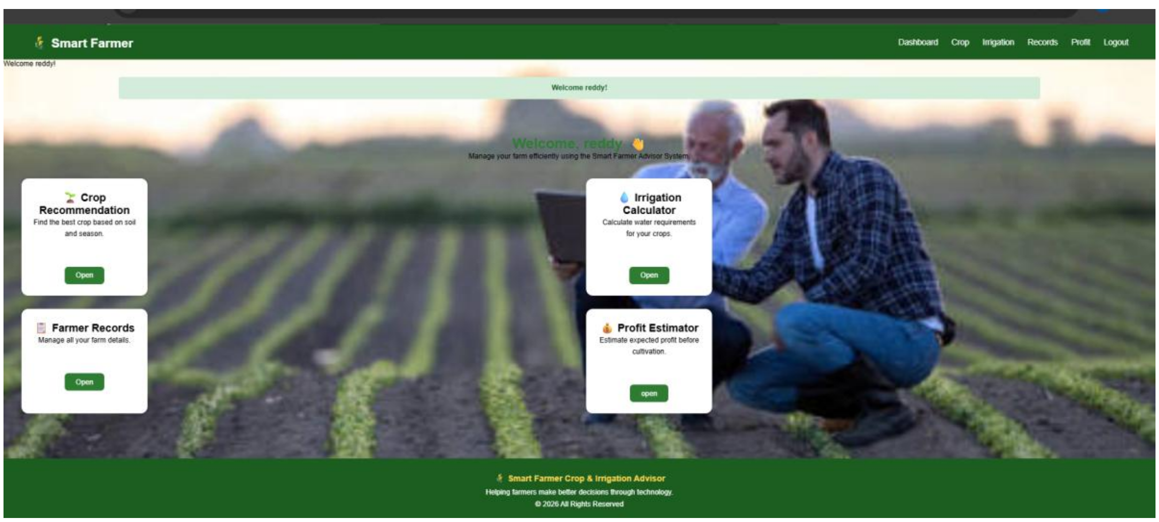
12.4 Dashboard Module

The Dashboard is the central module of the Smart Farmer Crop & Irrigation Advisor System. After successful login, users are redirected to the dashboard, where they can access all major features of the application. The dashboard acts as the control panel and provides quick navigation to Crop Recommendation, Irrigation Calculator, Farmer Records, and Profit Estimator.

The dashboard is designed with a clean and responsive interface to improve user experience. Each module is represented by a separate card with a brief description and an "Open" button. The navigation bar also provides a logout option to securely end the user's session.

Features

- Displays a personalized welcome message.
- Easy navigation to all modules.
- Responsive card-based layout.
- Secure logout functionality.
- Simple and user-friendly interface.



12.5 Crop Recommendation Module

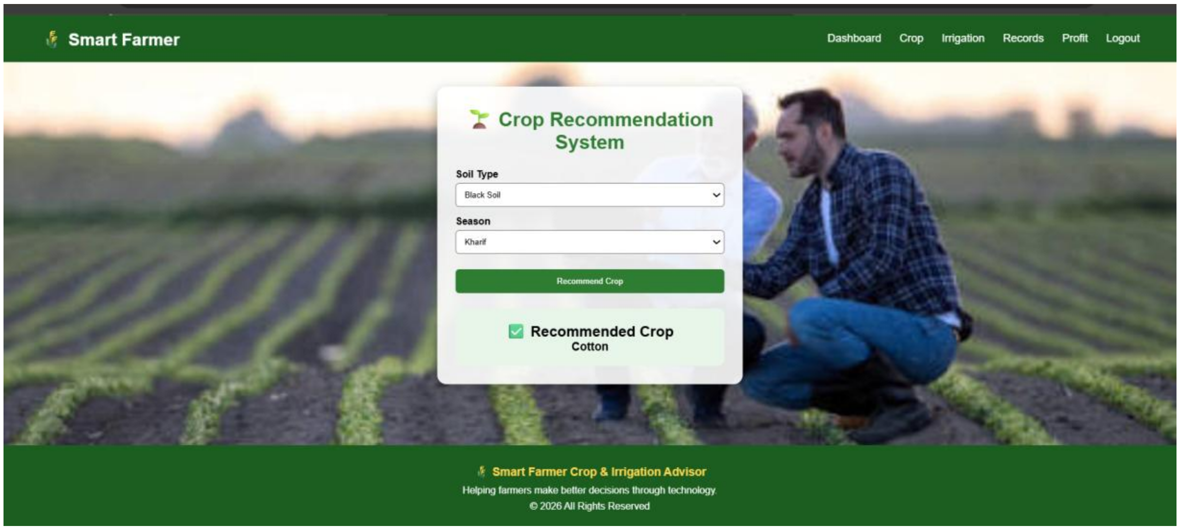
The Crop Recommendation module assists users in selecting a suitable crop based on the soil type and season entered by the user. The module follows a rule-based approach where predefined conditions are used to recommend the most appropriate crop.

The user selects the soil type (Black, Red, Clay, or Sandy) and the season (Kharif, Rabi, or Summer). After clicking the **Recommend Crop** button, the system processes the information and displays the recommended crop.

This module demonstrates how agricultural knowledge can be incorporated into a web application to support decision-making.

Features

- Soil type selection.
- Season selection.
- Instant crop recommendation.
- Simple and easy-to-use interface.



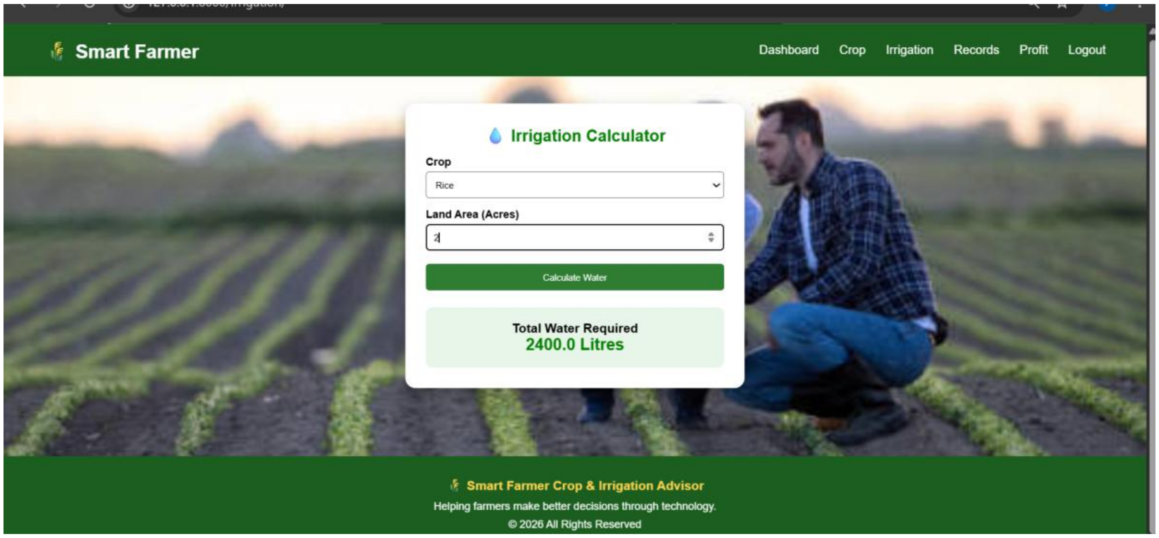
12.6 Irrigation Calculator Module

The Irrigation Calculator helps farmers estimate the amount of water required for irrigation. The user selects the crop type and enters the land area in acres. Based on predefined calculations, the system estimates the required amount of water and displays the result.

This module provides a quick estimation, helping users plan irrigation more effectively.

Features

- Crop selection.
- Land area input.
- Automatic water requirement calculation.
- Quick result display.



12.7 Farmer Records Module

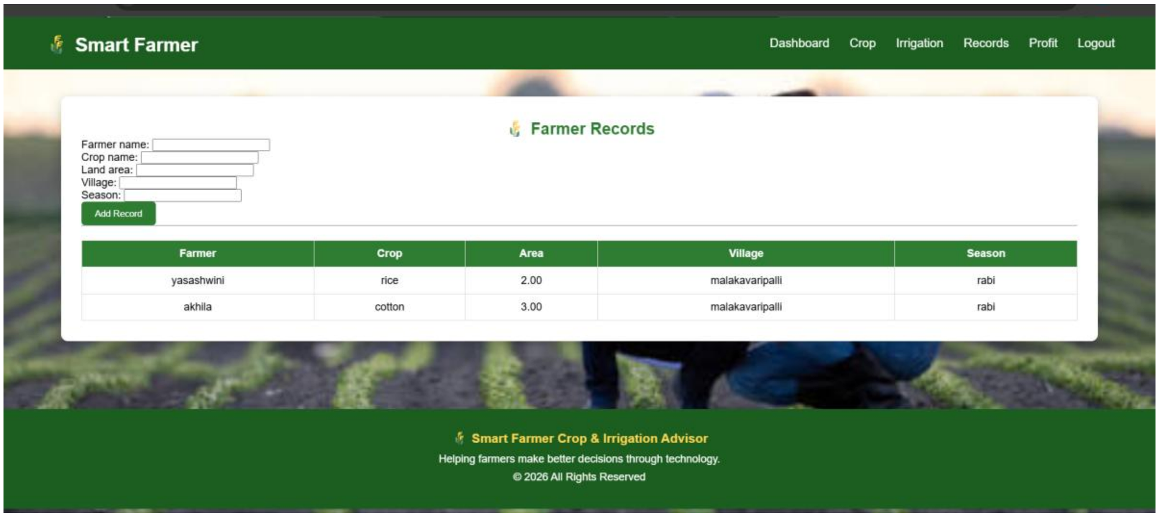
The Farmer Records module is responsible for storing and managing farmer information digitally. Users can add new farmer records by entering details such as farmer name, crop name, land area, village, and season.

All records are stored in the SQLite database and displayed in a structured table. This module demonstrates Create and Read operations (CRUD functionality) using Django.

Digital record management reduces paperwork and makes information retrieval much easier.

Features

- Add new farmer records.
- Store records in SQLite.
- Display records in tabular format.
- Easy data management.



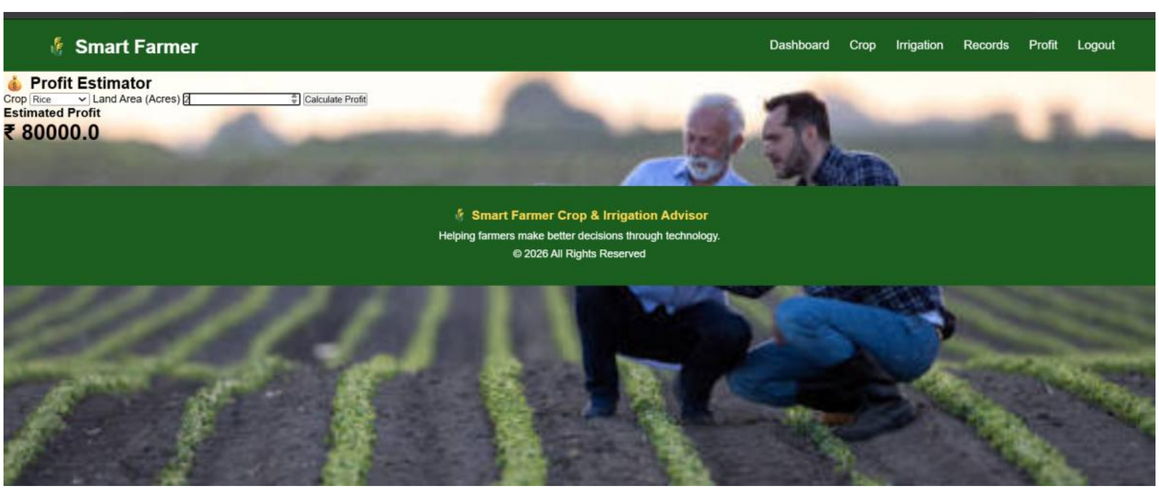
12.8 Profit Estimator Module

The Profit Estimator module helps farmers estimate the expected profit from cultivation. The user selects a crop and enters the land area. Based on predefined values, the system calculates the approximate profit and displays the estimated amount.

Although the calculation is based on fixed values for demonstration purposes, the module shows how agricultural data can be used to assist financial planning.

Features

- Crop selection.
- Land area input.
- Automatic profit estimation.
- Instant result display.

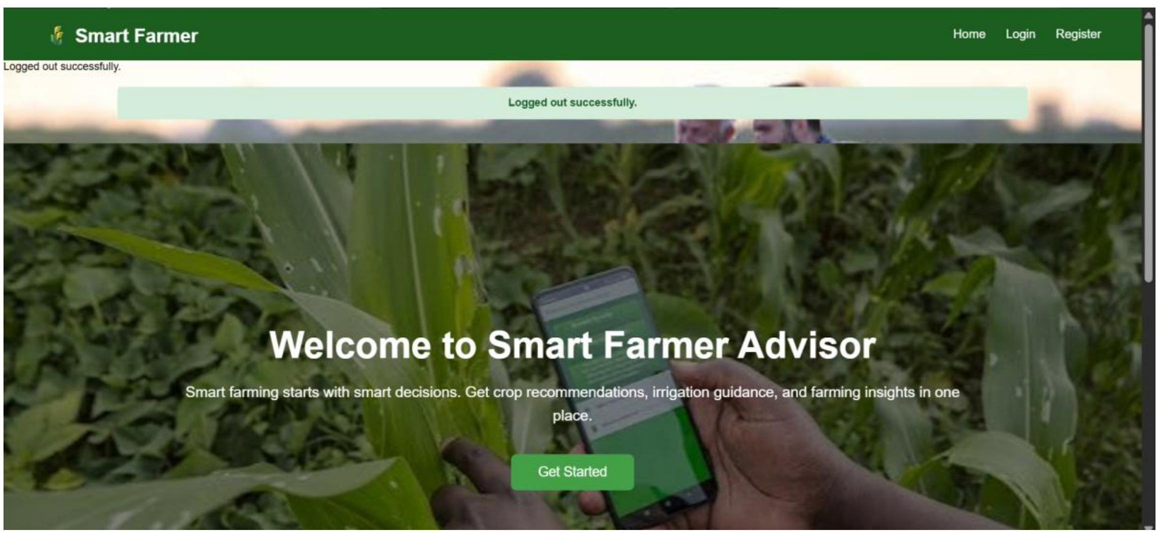


12.9 Logout Module

The **Logout Module** allows authenticated users to securely end their session after completing their work. When the user clicks the **Logout** button available in the navigation bar or dashboard, Django terminates the current session and redirects the user to the Home page.

This module enhances the security of the application by preventing unauthorized access after a user has finished using the system. It is implemented using Django's built-in authentication system, which safely removes the user's session information.

The logout functionality ensures that users must log in again to access protected pages such as the Dashboard, Crop Recommendation, Irrigation Calculator, Farmer Records, and Profit Estimator.



13. Database Design

The Smart Farmer Crop & Irrigation Advisor System uses **SQLite** as its database. SQLite is lightweight, easy to configure, and integrates seamlessly with the Django framework.

The database stores user information, farmer profiles, and farmer records. Django's Object Relational Mapping (ORM) is used to interact with the database, eliminating the need for complex SQL queries in most operations.

Main Tables

User Table

Stores authentication details such as:

- Username
- Email
- Password

Farmer Profile Table

Stores additional farmer information such as:

- Mobile Number
- Village
- Address (if applicable)

Farmer Record Table

Stores farming-related data:

- Farmer Name
- Crop Name
- Land Area
- Village

14. Advantages

The Smart Farmer Crop & Irrigation Advisor System provides several advantages over traditional methods of farm management.

Advantages

- User-friendly web interface.
- Secure user authentication.
- Easy crop recommendation.
- Quick irrigation estimation.
- Digital farmer record management.
- Accurate profit estimation.
- Reduced paperwork.
- Easy data retrieval.
- Lightweight SQLite database.
- Simple and responsive design.

15. Limitations

Although the project is functional, it has certain limitations.

Limitations

- Crop recommendations are based on predefined rules.
- Live weather information is not integrated.
- The application currently supports a limited number of crops.
- Profit calculation uses fixed values for demonstration purposes.
- SQLite is suitable for small-scale projects but not large enterprise applications.

16. Future Scope

The project can be enhanced further by integrating advanced technologies to provide more intelligent agricultural support.

Future Enhancements

- Integration of live weather forecasting using weather APIs.
- AI and Machine Learning based crop recommendation.
- Crop disease detection using image processing.
- Fertilizer recommendation system.
- Soil health analysis.
- GPS-based farm location services.
- Mobile application development.
- SMS and email notification services.
- Multi-language support.
- Cloud database integration.

17. Conclusion

The **Smart Farmer Crop & Irrigation Advisor System** was successfully developed using the Django Framework as a Python Full Stack web application. The system provides a simple and effective solution for managing agricultural activities through digital technology.

The application allows users to register securely, log in, receive crop recommendations, calculate irrigation requirements, manage farmer records, and estimate crop profits. The use of Django's MVT architecture ensures proper organization of the application, while SQLite provides efficient data storage.

The project demonstrates practical knowledge of Python, Django, HTML, CSS, JavaScript, and database management. It highlights how software solutions can simplify agricultural management and improve productivity.

Overall, the project meets its objectives and serves as a good example of a full-stack web application developed for agricultural assistance. It also provides a strong foundation for future enhancements such as AI-based recommendations, live weather integration, and mobile application support.

20. References

1. Django Software Foundation. *Django Documentation*. https://docs.djangoproject.com/
2. Python Software Foundation. *Python Documentation*. https://docs.python.org/
3. SQLite Documentation. https://www.sqlite.org/docs.html
4. Mozilla Developer Network (MDN). *HTML, CSS and JavaScript Documentation*. https://developer.mozilla.org/
5. W3Schools. *HTML, CSS, JavaScript and Django Tutorials*. https://www.w3schools.com/
6. Visual Studio Code Documentation. https://code.visualstudio.com/docs